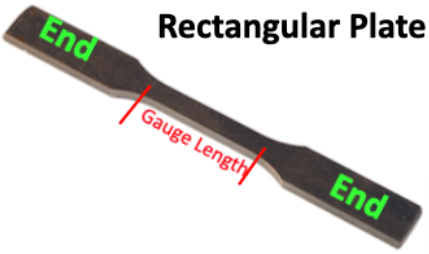
# Properties and Testing Methods

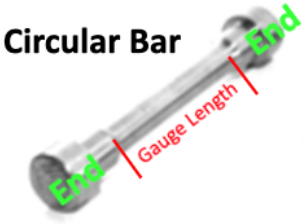
▼ Tensile Tests

tensile test samples

to do a tensile test, you need a dogbone tensile test sample, either with a rectangular or circular cross-section

the sample has two ends with a narrower middle, known as the gauge length $(L_0)$




The sample must follow the specifications:

$$L_0 = 5.65\sqrt{A_0}, \text{ for rectangular plates, } A_0 \text{ is the cross-sectional area}$$

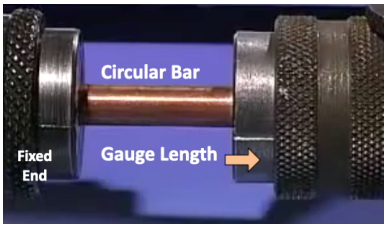
$$L_0 = 5d_0, \text{ for circular bars, where } d_0 \text{ is the diameter}$$

note that $A_0$ refers to original cross sectional area, and for circular bars,

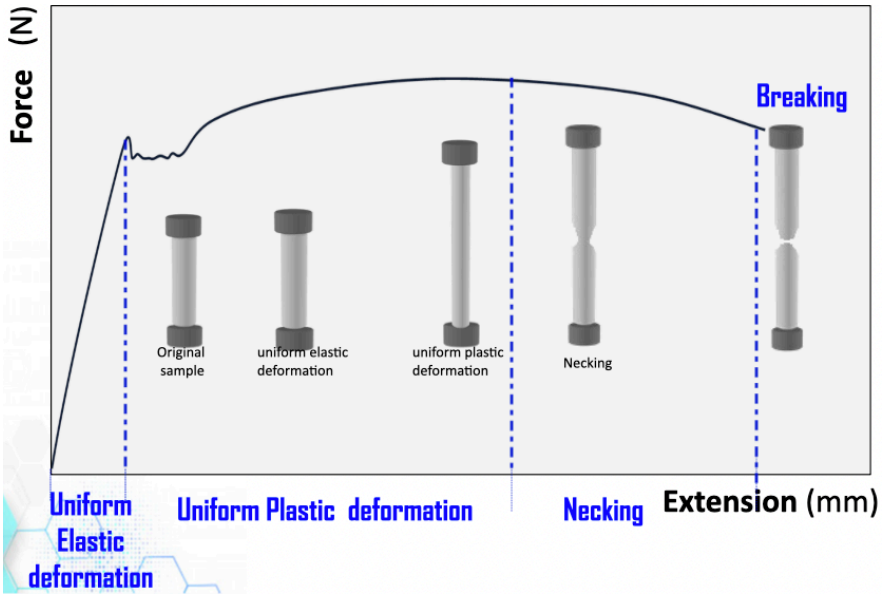
$$A_0 = \frac{\pi d_0^2}{4}$$

tensile test

The tensile test is conducted by subjecting the sample to a tensile force, where one end is fixed while the other end moves apart, increasing the tensile force until the sample breaks




force extension curve




the curve shows how much force is needed to extend the material to a certain point.

Uniform Elastic deformation

- force needed to extend material is directly proportional
- deformation is non-permanent, recoverable, up to a limit (elastic limit; highest force without permanent deformation)

Uniform Plastic Deformation

- past elastic limit, material undergoes UNIFORM plastic defomration. (uniform means that the material has the same thickness throughout)
- force experienced by material drops first and oscillates around a yield point (starting point of plastic deformation
- plastic deformation is permanent; non-recovrable
- this occurs up to a maximum point, which is the maximum amount of force that can be applied w/o fracturing

Necking

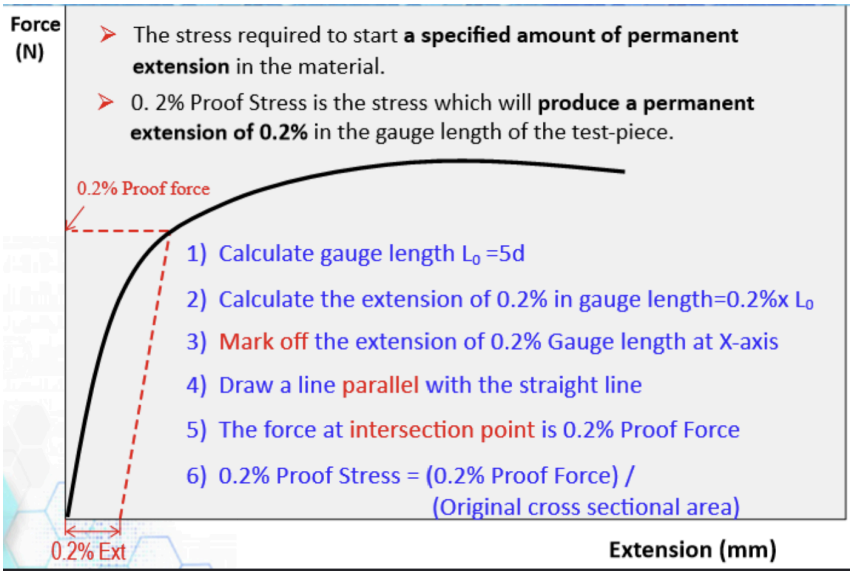
- past the max point, the sample does not undergo uniform defomration and thins out around a section (necking), which leads to fracture.

From the force extension curve, these quantities can be found:

- Ultimate tensile strength: $UTS = \frac{F_{max}}{A_0}$, determines amount of force sample can resist before rupture. (units is $N/mm^2$)
- Yield stress = $\frac{F_y \text{ (yield point force)}}{A_0}$, determines amount of force sample can resist before elastic deformation (units is $N/mm^2$)
- %E & %A: percentage change in the length after fracture !!!!(percentage elongation) and cross-sectional area respectively (formula is just percentage change), measures degree of plastic deformation & ducitility

Proof stress

- proof stress is used when the material does not have a yield point, i.e force extension curve is smooth, usually 0.2% proof stress. x% proof stress refers to the stress needed to leave a permanent extension of x% in the gauge length on the material




to find x% proof strength:

find gauge length, take x% of it, draw line parallel to straight line portion from the x%legnth point, find force at intesrection, find stress (force/A0)

Youngs modulus E

Youngs modulus is the ratio of stress to strain during elastic deformation, i.e. gradient of straight line portion on a stress/strain graph. (usually at a given point along the straight line portion)

$$\text{strain} = \frac{E, \text{extension}}{L_0}$$

$$\text{Youngs modulus} = \frac{\text{stress}}{\text{strain}}$$

strain has no units, as it is a proportion of how much the material is extended

the higher youngs modulus is/the steeper the straight line portion of the graph is, the more stiff the material is during elastic deformation.

▼ Hardness & Impact Tests

Hardness

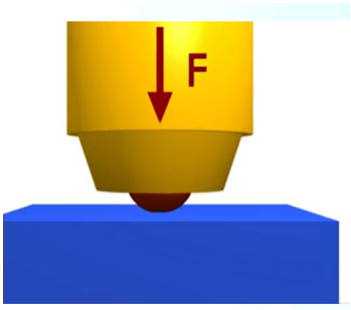
hardness is the measurement of how resistant a material is to abrasion, indentation or plastic deformation.

hardness tests help to quanitatively measure this. There are 3 main hardness tests, Brinell, Vickers and Rockwell.

Brinell Hardness test

The brinell test uses a hardened steel ball indenter to indent the material.

these indenters have a steel ball of diameter D, in standard sizes of 2.5mm, 5mm and 10mm.



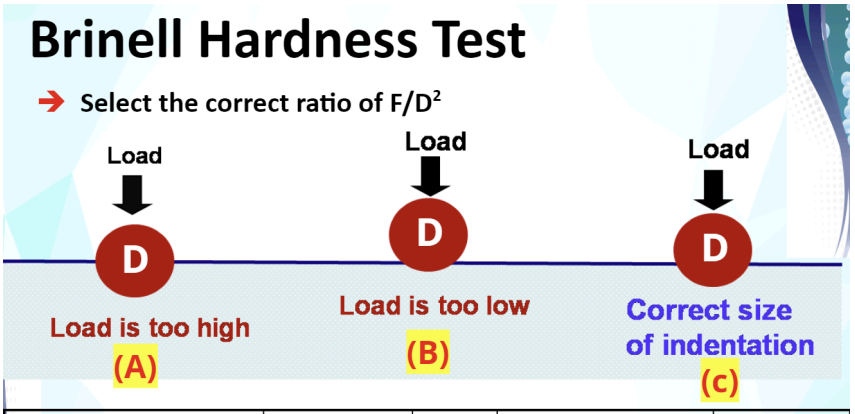
the indenter is pressed into the material at a fixed force F into the material, making a dent in the material where the indentation's diameter is measured as d.

The hardness of the material called HBN (hardness brinells number??) is a function of F, D and d. Usually HBN is found with a conversion table or is displayed by the testing machine. However the formula is as so:

$$HBN = \frac{F}{0.5\pi D\left(D - \sqrt{D^2 - d^2}\right)}$$

For the hardness test to be carried out, the thickness of the material needs to be thick enough to allow free unrestricted deformation of the material below the indentation and for the table of the testing machine to not affect the indentation and deformation.

thus, for soft materials, the thickness should be at least 15x the depth of impression and 7x for hard materials




| Materials | HBN | $F/D^2$ | Indenter Diameter (mm) | Load (kgf) |
|---|---|---|---|---|
| Steel & Cast Iron | Above 100 | 30 | 2.5 | 187.5 |
| Copper and Al alloys, | 30-200 | 10 | 2.5 | 62.5 |

The ratio of $\frac{F}{D^2}$ is also important to get accurate results. When F/D^2 is too high (because of too high of a load for the indenters diameter), the spherical indenter will sink all the way to the full diameter, making it useless (as D=d) [A]

Too low of F/D^2 will be inaccurate as well [B]

Thus, for different materials, differnet (standardised) values of F/D^2 is needed so that the indentation is significant enough to be measured but not too high to be equal to D. [C]

For steel, iron and other hard materials (HBN>100) need to have an F/D^2 of30.

Softer materials like copper and aluminium (HBN 30-200) need to have an F/D^2 of 10.

Given the indenter diamater or the load force, you can calculate the other values needed to test on a material.

Advantages:

Brinell scale closely relates to tensile strength (3*HBN=UTS)

Disadvatages:

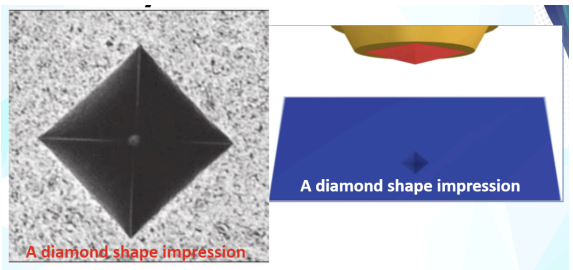
Cannot be used for very hard materials as they deform the indenter (lol)

Vickers Pyramid Hardness Test

Instead of a sphere indenter, a square base diamond pyramid (of apex angle 136* between opposite faces) indenter is used.

Similarly to Brinnels test, the indenter is forced into the material at a defined force and the impression is observed.

The diagonals of the diamond impression is measured as d1 and d2(measured under a microscope, indentation is very small). d is then calculated by taking the mean of the two values.




The hardness value, H.V. is dependent on F and d, as so:

$$HV = \frac{F}{d^2} = \frac{F}{\left(\frac{d_1+d_2}{2}\right)^2}$$

however HV is also more commonly found from a table/displayed by machine

Advantages:

Suitable for precision measurement

Accuracy is not dependent on depth of impression (as pyramid makes geometrically similar indents)

Useful for measuring very hard materials (but both hard and soft work, change load simialrly to brinells)

Negligible surface damage to material due to small indentation.

Disadvantages:

expensive because of diamond

more surface prep work needed

Rockwell Hardness Test

Rockwell hardness value is measured by the depth of impression into the material.

The test has 2 loads, the minor load and major load. for all 3 scales of the rockwell test, the minor load is 10kgf (kilogram force, amount of force exerted by 1kg).

The minor load is applied first onto the material by an indenter, to set a reference point to take the depth of impression, and to eliminate errors due to the surface condition.

the major load is applied afterward, and the difference in depth for both loads is used as a measure of hardness.

There are 3 scales used for rockwell test, A, B and C (labelled HRA, HRB, HRC respectively)

Indenter

Scale B uses a 1.6mm hardened steel ball, similar to Brinell, however with a different diameter

Scales A and C use a 120 degree Apex angle diamond cone, known as a brale. similar to Vickers, however with a lesser apex angle and with a cone

Major Load

B:100 kgf, C: 150 kgf, A: 60kgf

Uses

As scale B uses a steel ball, it works with softer materials like copper & aluminium alloys, and normalized steel

Scale C uses a diamond cone and a higher major load, so it is useful for thicker and harder materials like hardened steel.

Scale A uses the same indenter as C, however with a lesser major load, works better with hard and thin materials, like cemented carbides.

Generally, C and A are better for harder materials.

| Scale | Minor load (Kgf) | Major load (Kgf) | Indenter | Applications |
|---|---|---|---|---|
| B | 10 | 100 | 1.6 mm hardened steel ball | Copper Alloys/ Normalized Steel/ Aluminium Alloys |
| C | 10 | 150 | 120° Apex angle diamond cone or Brale | Hard materials/ Hardened Steel |
| A | 10 | 60 | | Hard and Thin / Shallow case / Cemented carbides |

Advantages:

Rapid in action

Suitable for both hard and soft materials ⟶ wider range of scale

Disadvantages:

too many scalres

does not work for very thin sheets and foils

not as realiable as vickers (in a lab setting)

scale does not relate to strength of metal used unlike Brinells.

Impact & Toughness

Toughness is the ability for a material to undergo shock or impact load (sudden force)

Impact tests help to measure toughness, by quantifying the energy required to break the test piece.

Impact Test(s)

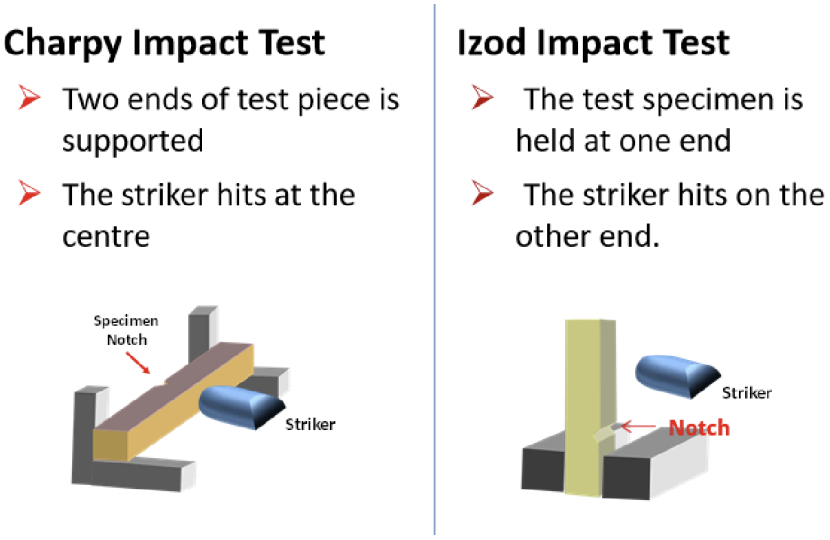
Generally, the impact test works by testing on a sample with a notch, to encourage the material to fracture at that point during impact.

The sample is then placed between a pendulum which strikes the sample. The difference between the final and initial position of the pendulum. The difference is usually measured on a scale connected to the pendulum. The higher the differnece, the more energy it can absorb under shock, the tougher it is.

There are 2 types of impact tests, both using a pendulum to strike the material.

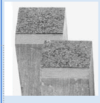
Charpy Test: the test piece is supported on both ends and the striker hits the center

Izod Test: the test piece is supported on one hand and the striker hits the other end.



During the test, the sample will change depending on its toughness.

Tougher materials will deform under impact while brittle materials will not deform/barely deform. Tougher materials will also have a dull crosssection after the impact while brittle materials will have a flat cross section.

| | Tough | Brittle |
|---|---|---|
| Energy absorbed (J) | 65 | 12 |
| Fracture surface/ characteristics/lateral deformation | Fracture Surface Characteristics: Fibrous/dull; Lateral deformation: Deformed | Fracture Surface  Characteristics: Crystalline/Flat; Lateral deformation: No / little Deformation |

Tougher metals will absorb more energy under impact, thus absorbed energy is used to deform the metal. Likewise, less energy absorbed by brittle metals means the

- Mechanical Properties

  Ductility

  Ability for a material to undergo plastic deformation due to TENSILE stress before fracture

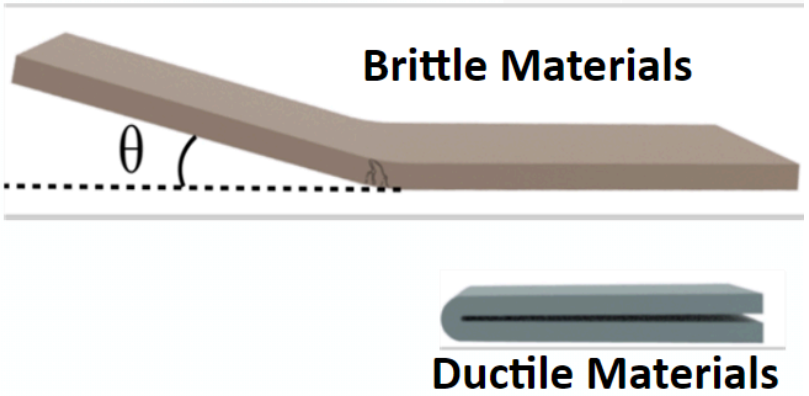
  Measured by %E and %A from tensile test

  Other tests to measure ductility:

  Bend test

  The plate is bent by an angle as much as it can without cracking.

  the higher it can bend (the higher the bent angle is), the more ductile it is

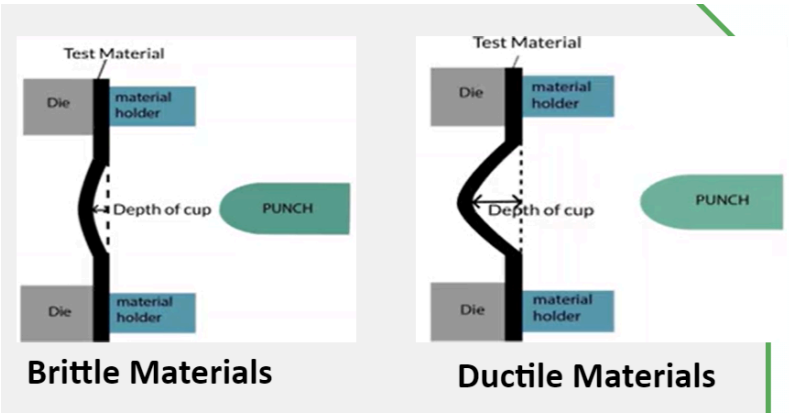



  Cupping Test

  Cupping test consists of a spherical indenter forced into a thin sample of a material.

  The sample is then stretched, forming a cup due to the indentor.

  The higher the depth of the cup, the more ductile it is.




Cold Working

Cold working is a type of metal working process by applying forces to plastically deform the metal below the recrystallisation temp.

**recrystallisation refers to the grains in the metal "re-organising themselves", usually in a stronger and less ductile formation.

working the metal under the recrystallisation temp (usually room temp) makes it less ductile, thus the metal needs to be VERY ductile to account for this

Cold working is used for processes like wire drawing and deep drawing (deep drawing process is similar to cupping test). both processes STRETCH the metal, putting it under tensile stress.

Metal needs to be very ductile (as said)

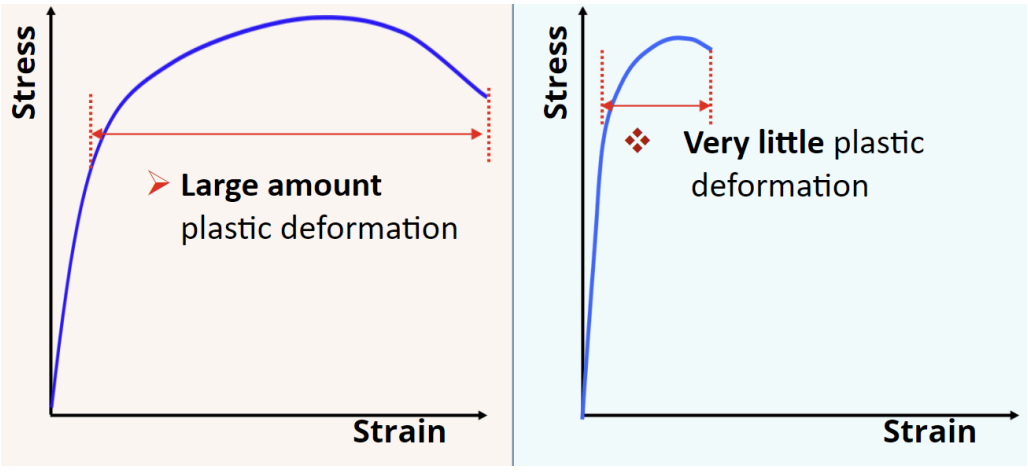
Stress applied needs to be greater than yield stress but less than UTS (to ensure plastic deformation and not elastic deformation or necking)

Malleability

Similarly to ductility, malleability is the ability for the material to plastically deform under COMPRESSIVE stress before fracture.

High Malleability is useful for stamping, forging and rolling, where the metal is COMPRESSED.

As both ductility and malleability are about plastic deformation, the larger the elastic deformation portion of the graph, the more it can plastically deform, making it more ductile/malleable.



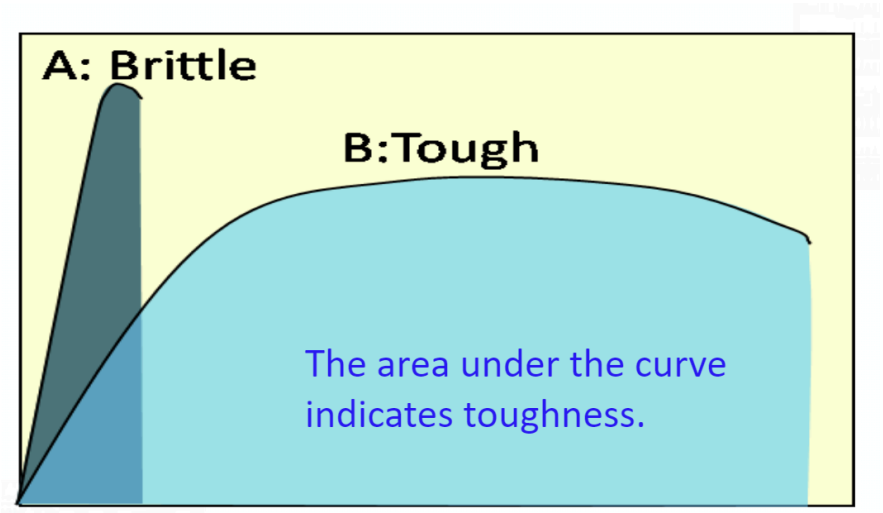

Toughness

Ability to undergo shock or impact load

Other than the impact test, toughness is also indicated by the area under the force-extension curve curve.

(area under curve units = force[(k)N] * distance[mm] = work done[J] As toughness is in joules, area represents toughness)

higher area ⟶ higher toughness




Hardness

Ability to resist indentation, abrasion and deformation

Hardness is related to strength (recall HBN = 3UTS)

(therefore hardness is indicated by the maximum of the curve)

harder materials will also be more resistant to wear , making it useful for cutting tools (e.g. in machining)

however, harder materials are harder to machine (due to being more resistant to deformation) (making in less machinable/low machineability

Harder materials are also less tough and more brittle (as they do not absorb  energy to deform)

this means that they have low ductility as well (as they cannot plastically deform)

These mechanical properties can be deduced from a stress strain curve of a material.

Strength: indicated by UTS,proof/yield stress (found from max point & yield point)

Hardness: indicated by UTS, higher UTS = higher hardness (HBN = 3UTS)

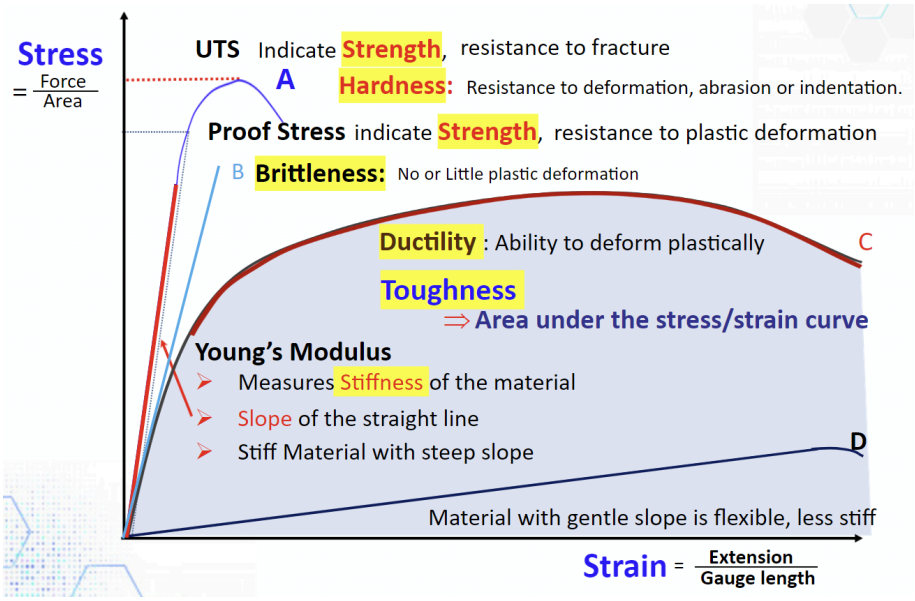
Stiffness: Youngs Modulus E; slope of straight line portion. Higher slope ⟶ higher stiffness

Ductility & Malleability: Plastic deformation portion of graph, aka horizontal difference between yield point and max point. higher difference ⟶ plastically deforms more without fracturing ⟶ higher ductility & malleability

Toughness & Brittleness:

Area under curve; higher area ⟶ more energy absorbed ⟶ tougher / less brittle (vice versa)

Summarised here:




likewise, when tables are used to list different quantities measured, we can use those to determine mechanical properties.

| | |
|---|---|
| Strength | UTS, proof stress, yield stress |
| Hardness | HBN, HV, HRA/HRB/HRC |
| Ductility | %E, %A |
| Toughness | impact energy |
| Stiffness | Youngs modulus E |
| idk | |

Fatigue

When metals are subjected to repetitive and alternating stresses, they may fracture even though the stresses are below its UTS. this is called fatigue failure

Fatigue is progressive, aka it increases and varies with time (as the component undergoes repetitive and cyclic stress, like rotational components). This also means that the stress is NOT constant over the period of time

Fatigue is unavoidable for all materials.

Creep

Similarly to fatigue, creep is also about stress over a period of time, however the stress is constant.

Creep is also about progressive plastic deformation, unlike fatigue, which causes failure (with no plastic deformation?)

Creep is dependent on time, load and temp.

when the temperature of the metal is higher than 1/3 of its melting point, progressive plastic deformation occurs, even when stress is less than UTS.

Creep is important in components under high temperatures, like turbines and boilers.

Summarising all the testing methods:

| | Tensile Test | Brinells (Soft) | Brinells (Hard) | Vickers | Rockwell B | Rockwell C | Rockwel |
|---|---|---|---|---|---|---|---|
| Load Amount/Type | Tensile Load | F/D^2 ratio of 10 | F/D^2 ratio of 30 | fixed amount of force | 10kgf minor load, 100kgf major | 10kgf minor load, 150kgf major | 10kgf mi load, 60 major |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Load Applier/Indenter | some pulling thing idk | Hardened Steel ball indenter (2.5mm, 5mm, 10mm) | Steel ball indenter (2.5mm, 5mm, 10mm) | Square base diamond pyramid (136 deg apex) | 1.6mm hardened steel ball | 120 deg apex diamond cone (Brale) | 120 deg diamond (Brale) |
| Measured Amount | Force needed to create extension | measured as function of force and diameter | measured as function of force and diameter | mean of the indentation's diagonals | difference in indentation depth of minor and major load | difference in indentation depth of minor and major load | differenc indentat depth of and maj |
| Sample Type (Metals/Sample prep) | dog bone sample | copper, aluminium | steel/cast iron | very hard materials | softer materials, copper, aluminium, normalised steels | hard materials, hardened steel | hard and materials cemente carbides |
| Advantages | | Brinell scale closely relates to tensile strength (3*HBN=UTS) | Brinell scale closely relates to tensile strength (3*HBN=UTS) | Accuracy is not dependent on depth of impression (as pyramid makes geometrically similar indents) Useful for measuring very hard materials Negligible surface damage to material due to small indentation. | Rapid in action Suitable for both hard and soft materials (wider range of scale) | Rapid in action Suitable for both hard and soft materials (wider range of scale) | Rapid in Suitable both har soft mat (wider ra scale) |
| Disadvantages | | Very hard materials will deform the indenter | Very hard materials will deform the indenter | expensive because of diamond more surface prep work needed | too many scales doesnt work for very thin sheets /foils not as reliable as vickers (in a lab setting) scale does not relate to strength of metal used | too many scales doesnt work for very thin sheets /foils not as reliable as vickers (in a lab setting) scale does not relate to strength of metal used | too many scales doesnt w very thin sheets /f not as re as vicke lab settin scale do relate to strength metal us |